ELECTRIC CHARGE & FIELDS

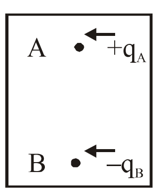
1. Two identical metal spheres A & B are charged with equal amount. If A is charged positively & B charged nagatively, then
(1) Mass of A increase
(2) Mass of A decrease
(3) Mass of B decrease
(4) No effect on mass by charging a body

2. On any body charge is additive by nature, means -
(1) Total charge on a system is constant
(2) Proper signs have to be used while adding the charge in a system
(3) Charge are of two types
(4) Total charge on a body is always zero

3. Charge can increase or decrease only in units of e, define -
(1) Conservation of charge
(2) Quantization of charge
(3) Total charge is zero
(4) Additive nature of charge

4. If electron & proton are 1Å apart, start moving towards each other, due to mutual attraction. Estimate acceleration of electron approximately.
(1) 1.4×10^{19} m/s^2 (2) 2.5×10^{22} m/s^2
(3) 1×10^{9} m/s^2 (4) 2.3×10^{-8} m/s^2

5. In a glass box charge A is fix, charge B in equilibrium, For vertical displacement charge B is in -



(1) Stable equilibrium
(2) Unstable equilibrium
(3) Neutral equilibrium
(4) Depends on mass & charge

6. If θ is the angle between electric field E & the outward normal to the area element then Number of field lines passing through this area is -
(1) Equal to $E\Delta S$
(2) Proportional to $E\Delta S\cos\theta$
(3) equal to $E\Delta S\cos\theta$
(4) proportional to $E\Delta S(\sin\theta)$

7. If Gauss's law $\oint \vec{E}.\vec{ds} = 0$, we can conclude that -
(1) E must be zero
(2) E must be perpendicular to surface
(3) Total flux through the surface is zero
(4) Flux is only going in the surface

8. Gauss's Law is true if force due to charge varies with distance (r) as -
(1) Force proportional to distance
(2) Force follow inverse square law
(3) Force is proportional to square of distance
(4) Gauss's law is independent with relation between force & distance.

9. While taking off synthetic clothes, seeing a spark or hearing a crackle, are due to -
(1) Motion of ions through air
(2) Production of shock waves due to motion of electrons
(3) electric discharge
(4) Cannot be explained

10. Out of gravitational, electromagnetic, vander Waals, electrostatic and nuclear forces, which two are able to provide an attractive force between two neutrons?
(1) Electrostatic and gravitational
(2) Electrostatic and nuclear
(3) Gravitational and nuclear
(4) Some other forces like vander waals.

11. In general, metallic ropes are suspended from the carriers to the ground which take inflammable material. The reason is -
(1) Their speed is controlled
(2) To keep the gravity of the carrier nearer to the earth
(3) To keep the body of the carrier in contact with the Earth
(4) Nothing should be placed under the carrier

12. When a body is connected to Earth, electrons from the earth flow into the body. This means the body was initially :-
(1) Uncharged
(2) Charged positively
(3) Charged negatively
(4) an insulator

13. To charge a body by induction -

(1) Body to be charged, must be an insulator

(2) Body to be charged, must be a semi conductor

(3) Body to be charged, must be a conductor

(4) Any type of body can be charged by induction

14. A body A is being charged by another charged body B by induction process. Then, charge acquired by A depends on -

(1) Nature of material of A

(2) Distance between A and B

(3) Nature of medium separating A and B

(4) All of the above

15. Electric field of a system of charges does not depend on -

(1) Position of charges forming the system

(2) Distance of point (at which field is being observed) from the charges forming system

(3) Value of test charge used to find out the field

(4) Separation of charges forming the system

16. A electric field line is a space curve, means-

(1) Field lines are hyperbolic curves

(2) Field lines are two - dimensional curves

(3) Field lines are three - dimensional curves

(4) Field lines are straight lines.

17. Two field lines can never cross each other because :

(1) Field lines are closed curves

(2) Field lines repels each other

(3) Field lines crowded only near the charge

(4) Field has a unique direction at each point

18. A charged particle is free to move in an electric field. It will travel :-

(1) Always along a line of force

(2) Along a line of force, if its initial velocity is zero

(3) Along a line of force, if it has some initial velocity in the direction of an acute angle with the line of force

(4) None of the above

19. The charge on an electron was calculated by-

(1) Faraday (2) J.J Thomson

(3) Millikan (4) Einstein

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Ans.	2	2	2	2	2	2	3	2	3	3	3	2	4	4	3	3	4	2	3